

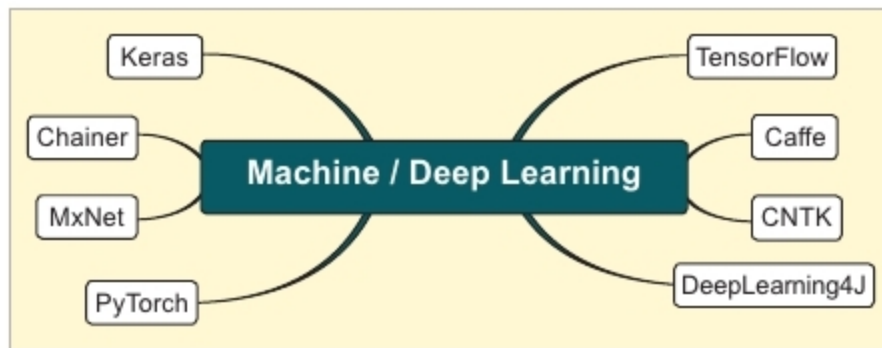


Figure 1: Machine and deep learning frameworks

Machine learning for mobile approaches

There are two different approaches to providing machine learning based solutions in mobile devices.

- **Option I: This keeps only the input and output interface on the mobile devices.** After getting the input, it is communicated to a server and the actual ML model functions there, and generates the output. The output is communicated back to the mobile devices through the selected output channel such as the display or speaker. For example, to perform image classification, the capturing of the image happens on the smartphone through its camera. The image is communicated to a server for the actual classification. The result of the classification is intimated to the user through the display or speakers.
- **Option II: No specialised servers to perform certain actions.** In this case, the actual ML model resides in the smartphone itself. So there isn't any need to send inputs or data to an external server. However, it should be noted that the training phase may need to be run on a powerful external system. The resultant model built based on the training phase is then shifted to the smartphone. Unlike the earlier example of image classification, in this option, there isn't any requirement for an external server during the actual classification phase.

Both the above approaches have their own advantages and disadvantages. Network delay may be an important barrier in Option I. The space and time complexity can be a barrier in Option II. However, with the exponential improvements in the hardware capabilities of mobile devices, these issues can be tackled without making the user realise any considerable lag. Many solutions have been built with real-time response features by adopting Option II.

Some of the popular problems that can be solved with mobile based ML are listed below:

- Speech recognition
- Image recognition/classification
- Object detection/identification
- Recognition of users' gestures
- Language translation

Mobile based frameworks and libraries

In Figure 1, the generic frameworks and libraries for ML are

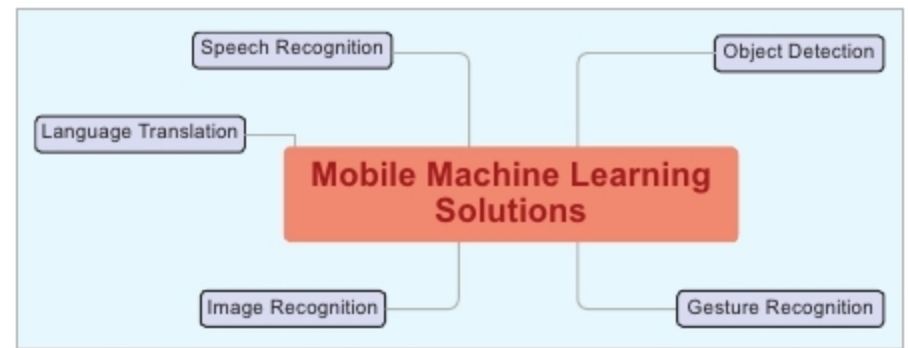


Figure 2: Mobile machine learning solutions

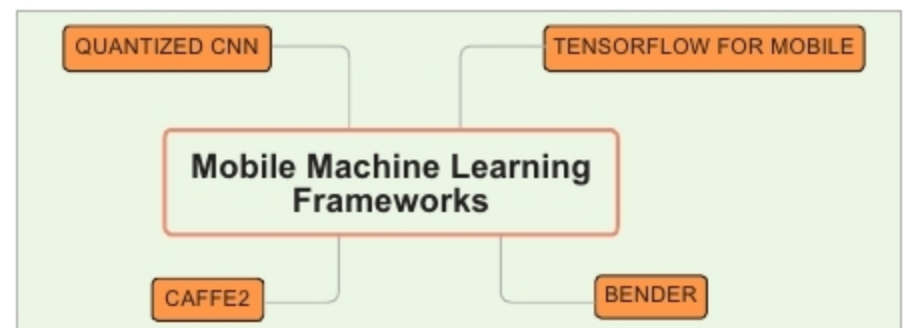


Figure 3: Machine learning frameworks for mobile environments

listed. However, there are certain specialised frameworks and libraries that support mobile environments. Some of these frameworks are listed in Figure 3.

This article introduces you to two such frameworks—TensorFlow and Bender.

TensorFlow

TensorFlow has evolved into a very popular and effective framework for implementing deep learning solutions. It offers two different variations for implementing deep learning on mobile and embedded devices:

- TensorFlow for Mobile
- TensorFlow Lite

TensorFlow Lite can be treated as a specialised porting of TensorFlow, focusing on devices such as smartphones. It leads to a smaller binary size and comparatively fewer dependencies. Due to these features, it yields a better performance.

Getting started: The first step in adopting mobile based ML is to confirm whether the problem can be solved effectively using machine learning on the mobile. The next steps are listed below:

- Build a data set with proper labelling
- Choose an optimal model for the problem at hand

Installing the required dependencies: The official documentation clearly lists the process involved in building an Android or iOS app with TensorFlow features (https://www.tensorflow.org/mobile/android_build).

- The first step is to install Android Studio. Detailed instructions on setting it up are available at <https://developer.android.com/studio/index.html>.
- The next step is to get TensorFlow from GitHub. This can be done with the following command:

```
git clone https://github.com/tensorflow/tensorflow
```